

LogitTrail: An Inspectable Local Interface for Typed Decisions from Frozen Language Models

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Abstract

A changed language-model decision can correct an error, move probability mass, or hide a stable but degenerate predictor. LogitTrail connects semantic candidate comparisons, panel-level outcomes, and source-record and physical-call provenance. A local API returns typed values from frozen next-token logits; a model-free browser exposes saved panels, while a separate importer compares users' physical records. Direct readout and native one-token decoding agree in all 1,350 matched pairs without an established latency advantage. Presentation and averaging studies use 22,000 calls: reduced sensitivity can accompany incorrect or nearly constant predictions. Further Japanese/English experiments do not establish added error-detection value from order sensitivity. These observations motivate displaying correctness, diversity, and cost alongside stability. Complete-JSON and ChainForge reconstructions retain equivalent diagnostic evidence. We contribute an auditable inspection workflow and scoped system evaluation; improved human performance remains untested.

1 Introduction

Applications may need a route, an urgency score, or a rubric stage. Typed interfaces make these contracts explicit, but valid types and concentrated candidate probabilities do not establish correct decisions.

LogitTrail (formerly Mini Jev) demonstrates this distinction through a local browser interface, an HTTP/Python API, and a saved-evidence explorer. Users can run typed questions locally or inspect recorded comparisons without downloading a model. The interface is inspired by TypeSafe's Jev (Almeida, 2026; TypeSafe, 2026b); this independent implementation does not reproduce its architecture, training, calibration, compatibility, or performance.

NLP developers comparing configurations need to determine which semantic probabilities changed, whether that corrected an error or hid a degenerate predictor, and which executions support the comparison. We connect aligned distributions, reference outcomes and panel frequencies, and source/call provenance to answer these questions. Complete JSON and custom ChainForge workflows retain equivalent evidence; evaluation establishes reconstruction and failure visibility, not faster human diagnosis. Candidate scoring and typed probability interfaces are established (Schick and Schütze, 2021; Ma et al., 2025; LMQL contributors, 2026); our contribution is their inspectable evidence path.

Source and artifacts: github.com/UpHash-Network/mini-jev. Browser access: [saved-evidence explorer](#). Video: [captioned demonstration](#).

2 System and Typed Contract

2.1 From a state to typed values

A request contains a shared state and named questions. Each question supplies a type, natural-language instructions, and candidate criteria. The service returns a semantic label, a candidate distribution, and a type-specific value: **Choice** returns the most likely candidate's key; keys are canonically sorted before prompting. **Noul** returns the true-candidate probability in a false/true decision. **Score** returns the expected zero-based rubric stage, alongside its most likely stage label. A Score expectation of 1.6 on stages 0–3 is a probability-weighted position, not a generated stage or calibrated utility. Interpretation assumes equally spaced stage indices.

Let z_i denote the next-token logit for candidate $i \in \{0, \dots, K - 1\}$, with K candidates and tem-

perature $T > 0$. The service computes

$$p_i = \frac{\exp(z_i/T)}{\sum_{j=0}^{K-1} \exp(z_j/T)},$$

$$s = \sum_{i=0}^{K-1} ip_i.$$
(1)

These are probabilities conditional on the selected candidate set. They are neither probabilities over all valid textual answers nor guarantees that the criteria are exhaustive. Changing the criteria changes the question as well as the normalization. The optional scalar temperature preserves the maximum-logit label, but can change Noul and Score values.

The interface also displays entropy concentration,

$$c = 1 - H(p)/\log K,$$

$$H(p) = - \sum_i p_i \log p_i.$$
(2)

A concentrated distribution has c near one. The display identifies this quantity as concentration rather than estimated accuracy. The response includes probability-semantics and calibration metadata, so downstream applications can retain the same distinction.

2.2 Local execution

The released inference configuration uses Qwen3.6-35B-A3B in Q4_K_M GGUF form (Qwen Team, 2026; ggml-org, 2026), through a source-built llama.cpp helper (ggml-org and contributors, 2026). The model and its output projection remain frozen; there is no trained decision head. The helper performs the full vocabulary projection and then gathers candidate logits. It does not implement a selected-row projection optimization.

The prompt repeats the task twice, following the use of repetition for non-reasoning inference (Leviathan et al., 2025). Choice and Noul use letter labels; Score with at most ten stages uses digits after a prefilled answer-object prefix. Larger Score rubrics use letters. The implementation checks that every candidate is a distinct single token at the actual prefix boundary. The API accepts 2–26 candidates; quality evaluation covers 2–8, while larger candidate sets have functional coverage only.

A multi-question request executes its questions sequentially. Model state is cleared between questions: they share the supplied state text, but not a

cached prefill. Complete inputs exceeding 2,048 tokens are rejected rather than silently truncated. Schema validation and deterministic response construction prevent free-form output from crossing the API boundary; these software properties do not prevent semantic errors.

2.3 Relationship to one-token decoding

Readout and candidate-masked one-token decoding have the same distribution when prefix, logits, candidate IDs, temperature, and processors match. Greedy labels agree except for tie-breaking; stochastic samples need not match. Choosing the first token after prefill need not require another forward pass, so avoiding generation does not itself imply a speedup. Section 4 tests an actual native sampler under matched conditions.

3 Demonstration and Access

3.1 Inspect saved evidence without a model

The static [saved-evidence explorer](#) requires only a browser. It contains all eligible records from the presentation and averaging studies: 22,000 physical calls and 9,000 derived answers in 18 panels. Their 3,600 study/item/model combinations represent 1,200 distinct source items, reused across checkpoints. These are recorded observations, not new inference.

Users choose a study, checkpoint, task, item, and two conditions. Both sides retain the same item and model; probabilities align by semantic key on a common 0–100% scale even when token bindings or row order differ. The display separates the selected label, typed value, concentration, and reference. Full-panel label frequencies remain visible independently of item selection. This permits inspection of an order flip, a correct answer lost after averaging, or a nearly constant predictor. Four outcome-selected examples illustrate these cases; they do not estimate prevalence.

The explorer also distinguishes executions from derived results. An averaged answer lists its physical members; comparisons count their union without double-counting shared calls. Source details identify archive members, JSONL lines, mappings, and hashes. Loaded panels are SHA-256 checked; comparison exports retain exact values and sources. Original task text is withheld. URL fragments share selections. Software checks match all exported records, 172,200 dependent comparison pairs, and 155 panel distributions

to raw calculations. A complete-JSON view retains the same information; these are not human-performance results.

A separate [local JSONL importer](#) aligns user-supplied physical logits or declared full probabilities and exports typed comparisons without uploads or storage. Missing references/latency remain unknown; source authenticity and derived-member recipes are unsupported. Synthetic tests and one hash-selected archived item check compatibility, not practical advantage.

3.2 Run local typed decisions

The separate live browser connects through a Python bridge to the local model API. Users edit a shared state, instructions, and criteria for Choice, Noul, and Score, then select *Run decisions*. Results expose candidate probabilities, concentration, elapsed time, and request/response JSON. Editing marks prior results stale. The live UI has no replay or cyclic-averaging mode: an unavailable API produces an error. Input edits explore behavior, not controlled accuracy.

A linked [reviewer guide](#) proceeds from the explorer to model-free verification and optional live installation. The tested live environment is a 64 GB Apple M5 Pro; its model file is approximately 20.4 GB. Appendix A gives installation and verification boundaries. The static explorer has no model, API-key, or build requirement.

4 Evidence Behind the Demonstration

4.1 Evaluation panels and provenance

Table 1 separates the evidence sources. Repeated requests are not additional quality examples, and the external tasks do not share one accuracy metric.

The local regression suite has 2,220 generated items from 45 template families and 180 individually AI-authored items; its legacy manual field does not mean human annotation. Earlier errors informed authoring. All 2,400 outputs were valid, but 162 labels disagreed with authored references. Temperature fitting did not uniformly improve calibration or score error.

The expanded sample has 200 public-development examples each from JCoLA (Someya et al., 2024), JSTS, and JCommonsenseQA (Kurihara et al., 2022). Hash selection and rubrics were frozen before inference, without independent preregistration. Six JSTS rows

Evidence	Size	Interpretation
Local regression	2,400 items	Self-authored Japanese; 800 per type
Expanded external	600 items	JCoLA, JSTS, JCQA; 200 each
Matched runtime	4,050 quests	150 local $\times$ 5 repetitions, plus 600 external; 3 modes
Presentation study	7,600 quests	Same 600 external items; 3 checkpoints
Cyclic averaging	14,400 requests	600 additional external items; 3 checkpoints
Diagnostic replay	2,400 quests	240 further JCQA items; 2 checkpoints
Scope extension	4,800 quests	240 new JCQA + 240 English items; 2 families

Table 1: Evaluation accounting. The original panels use 1,200 items; two later diagnostic studies add 240 and 480. Measured counts exclude synthetic checks and warm-ups. JCQA abbreviates JCommonsenseQA.

share text or image provenance with an earlier 300-item JNLI pilot. Public development data may have appeared in model training; these are not independent hidden-test evaluations.

4.2 Matched runtime and parity

All three conditions used one resident model and a common research HTTP response containing a type and semantic label. Direct readout and native one-token greedy selection used identical complete prompts and candidates. The JSON condition actually generated a grammar-constrained object, with a serialization-specific prompt. Its timing comparison therefore includes prompt and pre-fill differences as well as multi-token generation; it does not isolate sampler overhead.

All $(150 \times 5 + 600) \times 3 = 4,050$ requests completed without recorded failures; 21 synthetic warm-ups are excluded. Mode order was randomized within item/repetition. Research-harness timing spans serialization through loopback HTTP parsing, including audit transfer but excluding trace writes and browser rendering.

Direct and one-token conditions matched prompts, token sequences, candidate IDs, logits, and labels in all 1,350 paired observations. Maximum observed candidate-logit difference was zero; there were no maximum-logit ties. The primary latency statistic is the median across 150 local items of the within-item difference between five-repetition mean latencies. It is not a difference between marginal medians.

Table 2 uses 10,000 whole-family/tag percentile bootstraps, conditional on this model, machine,

Candidate comparison

[Open complete JSON for this selection](#) - Includes the selected comparison, full panel, and source index.

A Recorded condition

Original presentation

The saved original presentation.

B Recorded condition

Reversed candidate rows

Reverse only the displayed rows, preserving answer-token, key, and meaning bindings.

The most likely answer stays the same. Largest candidate probability change: 7.384769 percentage points.

Most likely semantic answer

A true

B true

Recorded tie handling is preserved.

True-candidate probability

A 0.986484

B 0.912636

Within the false/true candidate pair.

Concentration

A 0.8967

B 0.5724

0 = uniform · 1 = peaked; not accuracy.

Model calls per result

A 1

B 1

0 shared; 2 distinct calls across A and B.

Probability within the candidate set

A and B share a 0–100% scale. Δ is B – A in percentage points. Values are aligned by semantic key, even when presentation order changes.

false · unacceptable

key: false

A

B

1.351581%

8.73635%

+7.384769

Δ percentage points

true · acceptable

key: true

A

B

98.648419%

91.26365%

–7.384769

Δ percentage points

Dataset reference label: true. A matches; B matches. This is one recorded item, not a new accuracy estimate.

► Answer frequencies across this full panel

Concentration measures how peaked a candidate distribution is; it is not the probability of being correct. Score expectations assume equally spaced stages. Recorded examples do not establish human usability or a causal model-size effect.

Figure 1: The saved-evidence explorer aligns recorded candidate probabilities by meaning. Users inspect labels, typed values, references, and physical-call costs, then trace or export the comparison. The displayed example is an illustration, not a representative performance sample or a live model execution.

Comparator — direct	Difference	95% interval
One token	+0.0898	[–0.7483, 0.8449]
JSON	+160.2958	[138.1189, 177.0587]

Table 2: Primary paired HTTP latency differences in milliseconds. Intervals resample observed local family/tag clusters; JSON uses a different serialization prompt.

items, and session. The one-token interval crosses zero; JSON is slower under different serialization requirements. These intervals do not estimate between-machine or independent-session variability.

4.3 External task-specific results

Direct readout and one-token selection have identical external labels. JCoLA reached 86.0% accuracy and MCC 0.5708; JSON reached 82.0% and MCC 0.5160. The selected sample contains 158 acceptable sentences, so an always-acceptable rule already achieves 79.0% accuracy and has un-

defined MCC. Reporting only accuracy would obscure this imbalance.

JCommonsenseQA accuracy was 95.0% for direct/one-token and 95.5% for JSON, a difference of one example. JSTS retains its continuous reference scores in $[0, 5]$: direct expected-stage MAE at $T = 1$ was 0.5065. JSON has no comparable probability vector, so we do not assign it an expected-stage metric. The separate hard-stage comparison gives MAE 0.5480 for direct/one-token and 0.5650 for JSON. We neither round gold scores into classes nor pool task metrics.

4.4 Presentation sensitivity and averaging

Two subsequent studies add Qwen2.5-0.5B-Instruct and Qwen2.5-1.5B-Instruct to the native 35B checkpoint. The first uses 7,600 requests on the same 600 external items. All 1,800 exact-input repeat pairs have identical logits and probabilities, yet display reversal changes native JCQA labels on 11/200 items. Conversely, the 1.5B checkpoint has zero JCoLA flips while always predicting ac-

Checkpoint	Accuracy (%)	Flip rate (%)
Qwen2.5 0.5B	39.5 → 36.0	65.5 → 30.5
Qwen2.5 1.5B	80.0 → 83.5	26.0 → 12.5
Qwen3.6 35B-A3B	98.5 → 98.0	4.5 → 1.5

Table 3: JCQA on the additional 200-item panel: single → cyclic point estimates. Flip rate measures orientation disagreement, not error rate. Native 35B uses Q4_K_M. These are not equal-compute comparisons.

ceptable: stability alone is insufficient.

The second freezes 600 additional public-development items before its 14,400 requests. It excludes prior IDs and literal text reuse, and selects JSTS pairs from previously unused sentence/image components. However, 194/200 JCoLA items share bibliographic sources with earlier evaluations. Both studies complete without inference errors; each excludes 57 synthetic checks/warm-ups.

We average semantic candidate probabilities at $T = 1$ across cyclic rotations of complete option lines, keeping output-token-meaning bindings fixed. Fixed-binding transformations and cyclic aggregation have precedent (Wei et al., 2024; Zheng et al., 2024); this is not a new algorithm or a reproduction of PriDe. Table 3 compares canonical-single and forward-cyclic accuracy, alongside single/reversed and forward/reverse-cyclic label disagreement. Lower disagreement does not consistently improve accuracy. The 0.5B cyclic JCoLA output is always acceptable (78.5% accuracy, undefined MCC). On JSTS, 0.5B and 1.5B cyclic hard labels occupy one stage on 199/200 and 200/200 items, respectively. These marginals expose collapse hidden by type validity or stability.

One cyclic answer costs 2, 5, or 6 model calls for JCoLA, JCQA, or JSTS; the single baseline reuses a pool member. Binary forward/reverse pools contain the same two calls, so their equality is structural. There is no equal-call repeat control in this averaging study. The small checkpoints use Transformers/MPS float32; the native model uses quantized Metal inference. Architecture, precision, and backend differ, precluding a model-size or cross-backend speed conclusion. These are descriptive system checks; fuller distributions, grouped intervals, and call accounting accompany the artifacts.

Diagnostic value. A frozen 240-question JCQA panel adds 2,400 calls across Qwen 1.5B and 35B; adding order-induced total variation (TV) to entropy did not establish added error-detection value. A scope extension adds 4,800 calls on 240 new JCQA and 240 English CommonsenseQA items (Talmor et al., 2019), shared by Qwen 1.5B and Phi-4-mini (Microsoft, 2025). Its primary Phi-English AP difference is -0.051 ; at 720 calls, both policies return 174/240 correct answers. All intervals and budgets are retained (Appendix A). A post-hoc aggregation control on the same extension responses changes 4–8 labels per 240 items between arithmetic and geometric means, without consistent accuracy gains (Table 6). Sensitivity remains an inspection signal, not a confidence or routing guarantee.

4.5 Reproduction and framework integration

The original package retains 26,050 requests, protocols, hashes, and analyzers. Clean-directory CPU replay reproduced 56 derived files byte-for-byte. A fresh native build served all three types with a cached, hash-verified model on the same Mac: source/cache reconstruction, not independent-machine replication.

Synthetic integration checks retain failures. With matched checkpoint, prefixes, and candidate IDs, a custom LMQL 0.7.3 backend matched 12/12 labels, but three probability differences exceeded 10^{-5} (maximum 2.63×10^{-5}). ChainForge’s custom provider preserved eight typed-answer pairs and rejected three invalid inputs. These are interoperability checks (Appendix A), not human or quality evaluations. A retrospective engineering case also retains a failed numerical gate, its amendment and subsequent measurements; exact distributions reveal discrepancies hidden by matching labels.

5 Related Work and Scope

Candidate and ordinal scoring. Verbalizers (Schick and Schütze, 2021), decoding-free selection (Ma et al., 2025), Qwen3 reranking (Zhang et al., 2025; Qwen Team, 2025), and G-Eval’s weighted scores (Liu et al., 2023) precede our readouts. Expected ordinal relevance appears in zero-shot ranking (Zhuang et al., 2024) and trained job-candidate scoring (Wang et al., 2026). We claim neither a new scoring rule nor ranking gains.

Structured interfaces. LMQL scores continuations (LMQL contributors, 2026); Outlines, Guidance, and grammars constrain generation (dottxt and Outlines contributors, 2026; Guidance contributors, 2026; Geng et al., 2023). TypeSafe’s adapter returns generated rather than candidate-logit probabilities (TypeSafe, 2026a). Native llama.cpp source serves supported decision models, including LEV order averaging, not arbitrary GGUF models (Nguyen, 2026). LLM2Jev scores complete bracketed numeric suffixes on frozen backbones and separately studies fine-tuning (Li and Wagle, 2026); this differs from one-token readout. AnyJev studies label-prior correction and log-space cyclic aggregation with position-bound labels (Zhang et al., 2026); we retain fixed token-meaning bindings. OmniMed-Jev trains a medical multimodal candidate head with LoRA (Tang and Chen, 2026); AudioJev trains supervised audio order consistency (Lv et al., 2026). Our serialization and aggregation checks do not benchmark these systems.

Inspection interfaces. Rakutko’s distinct minijev compares frozen letter-logit scoring with constrained JSON in an inspection bench (Rakutko, 2026); our rename avoids confusion. Ollama documents local Choice/Noul/Score probabilities (Ollama, 2026); ChainForge adds an Ollama decision provider (Arawjo et al., 2024; ChainForge contributors, 2026). Its inference path remains untested here. ChainForge 0.3.7.6’s GUI retains six matched saved-evidence cases; its native Processor also executes our custom JavaScript to reconstruct 2,400 conditions from 6,000 archived logit records within 10^{-12} tolerance. This supports competitor feasibility, not inspection superiority (Appendix A). LLM Comparator compares responses (Kahng et al., 2025); Langfuse and Phoenix expose typed distributions or metadata (Langfuse, 2026; Arize AI and Phoenix contributors, 2026). New promptfoo source retains Jev answers in regular and rubric evaluation paths (Nobi, 2026); these source integrations are untested here. Our distinction concerns connecting complete panels, semantic alignment, shared calls, and archive-line provenance.

Sensitivity and calibration. Order/token bias and fixed-binding changes are established (Wei et al., 2024; Zheng et al., 2024). Preprints separate order stability from accuracy (Hanna and Feng, 2026), ordinal hard-label collapse from con-

centration (Gao et al., 2026), and correlated judge errors from confidence-cascade savings (Rao and Callison-Burch, 2026). SenSel uses bias-adjusted sensitivity to instruction or demonstration-order changes for abstention (Chen et al., 2023). Its perturbations and endpoint differ from our candidate-order TV; our nulls neither reproduce nor refute it. Temperature scaling (Guo et al., 2017) differs from label-bias correction (Zhao et al., 2021); canonical sorting does not establish invariance.

Limitations

Presentation studies cover three checkpoints on one Apple Silicon device; matched parity/latency covers one native quantized model in one session. Architecture, precision, and backend differences prevent model-size attribution. Public subsets have unknown training contamination and source dependence. The English extension covers one five-choice task; model sizes and languages are confounded with family and dataset. Local data are AI-authored and development-related, without independent human reference validation. No user study establishes usability or downstream decision quality.

Candidate probabilities vary with prompts, labels, coverage, and temperature. Concentrated wrong answers occur; type validity and entropy concentration cannot replace application-specific validation before consequential actions. Score assumes equally spaced stages; Noul is a relative candidate probability, not a verified truth frequency. Questions run sequentially on a large model; other operating systems and lower-memory configurations remain unvalidated. Live-demo timings are not directly comparable with the research harness’s controlled interval.

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Ethics and Availability

The demonstration supports model inspection and typed-interface prototyping, with no evidence for high-stakes deployment. Inference is local; users remain responsible for inputs and downstream use. Code and self-authored local data use MIT terms. Japanese-dataset-derived selection meta-data, rubrics, protocols, and results retain applicable CC BY-SA 4.0 terms; English CommonsenseQA material retains MIT terms. Upstream dataset text, model weights, and compiled binaries are not bundled; repository data cards and notices specify acquisition and attribution.

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A Comparison Scope and Evidence Access

Comparison	Shared conditions	Recorded observation	Unmeasured or different
Native one-token	Checkpoint, full prefix, candidate IDs/logits, greedy selection	1,350/1,350 labels match; paired latency interval includes zero	Other machines and independent sessions
Grammar JSON	Resident model, selected tasks, HTTP timing interval	Validity, task-specific quality and serialization latency	Prompt and generated length differ; no probability-vector parity
LMQL 0.7.3	Checkpoint, full prefix, candidate IDs; custom native backend	12/12 labels match; 3/12 probability comparisons exceed 10^{-5}	Native evaluation paths differ; not a stock-backend benchmark
ChainForge 0.3.7.4	Same local API, request content and typed-response contract	Eight answer pairs identical; three invalid inputs rejected	Browser flow editor and human performance
ChainForge 0.3.7.6 GUI	Six saved cases; separate raw-logit/recipe input with custom JavaScript	Saved exports agree; native Processor reconstructs 2,400 conditions within 10^{-12}	Custom diagnostic code; no inference or human performance
Other typed tools	Source only: Langfuse, Phoenix, promptfoo; native System One route	Typed evaluation, distribution meta-data and specialized-model serving	No matched inference, GUI trial or comparative user outcome

Table 4: Evidence tiers, not a capability ranking. Unmeasured properties are not absent capabilities. Framework diagnostics are separate from the task-quality panels.

Public and internal inspection. The released HTTP response and browser expose semantic labels, typed values, candidate probabilities, concentration, and probability-semantics metadata. Raw candidate token IDs and logits belong to the internal engine and research traces; they are not fields in the public response. Table 4 distinguishes controlled comparisons, integration checks, and source-based related work. These observations do not establish that people work faster or make better decisions with this interface.

Model-free evidence review. The frozen source/evidence ZIP retains the original 26,050 measured requests; 300 earlier pilot predictions are background and excluded. Standard-library checks verify accounting, and CPU replay reproduces 56 derived files without models or network access. The current manuscript and later studies are separate packages. The explorer validates unchanged source hashes, semantic mappings and physical-call unions. Reanalysis verifies calculations, not observation authenticity or generalization.

Live installation. The reviewer guide supplies native build, pinned-model acquisition, and `./run.sh --without-calibration` commands. Running `python3 -m demo.server --port 8766` serves the interface at <http://127.0.0.1:8766/>. Fresh builds use $T = 1$; historical calibration is runtime-bound. Tests use macOS 26.4, Metal, and Accelerate. The 20.4 GB model requires addi-

tional runtime memory; a minimum configuration is unestablished. We distribute source, not weights or binaries. Serving docs/ over HTTPS or local-host enables model-free browser hash verification.

Post-hoc LMQL diagnosis. All three original 10^{-5} probability failures remain. Maximum Score expectation error is 7.18×10^{-5} , within the separate 10^{-4} typed-value tolerance. Native probability margins exceed 0.9799, so matching labels provide no near-boundary evidence. Softmax of saved LMTP logits reproduces LMQL probabilities within 3.64×10^{-9} ; native-path logit differences can account for discrepancies, but their cause remains unisolated. Type, candidate count, and prefix are confounded. This analysis adds no inference and changes no threshold.

Matched saved-evidence workflow. The [saved-evidence package](#) freezes six ID-hash-selected cases and 32 questions before GUI execution; earlier outcomes were known. All presenters receive complete panels and source indices. ChainForge’s native search, lightboxes, spreadsheet and flow exports retain evidence supplied by the adapter, which computes alignment, deltas and call unions. No human performance was measured.

Reconstruction from raw records. The [raw-record package](#) instead supplies logits, mappings, temperatures, gold references and member recipes. ChainForge executes custom JavaScript from an empty cache in Firefox 153, reconstructing 2,400

conditions from 6,000 records. An independent Python oracle agrees within 1.78×10^{-15} . Six tuple-reversal controls preserve results; eight malformed inputs are rejected. These software checks establish competitor feasibility with custom code, not human advantage or new inference. The later physical-record importer does not ingest this fixture’s derived-member recipes.

Error detection and budgeted replay. The [original diagnostic package](#) separates inspected exploration from 240 new JCQA questions and 2,400 calls. The [scope extension](#) adds distinct Japanese/English 240-item panels and 4,800 calls across two families. Selection, prompts, models and analyses were locally frozen before benchmark inference; prior null results were known. Neither study is independent preregistration or a contamination-free test. The extension uses float32/MPS, stock full-position vocabulary projection and candidate gather for both models. Its Phi optimization gate failed before benchmark inference. Subsequent diagnosis found two of twelve synthetic pairs above the fixed 10^{-5} probability tolerance (maximum 1.28×10^{-5}), despite identical labels. A pre-benchmark amendment replaced last-position-only with stock full-position projection; repeat checks were identical, without proving parity with the rejected path. A [retrospective case](#) imports all 48 diagnostic/repeat records, preserving exact sources and complete JSON as a baseline. The original diagnosis was command-line based; no GUI discovery or accuracy repair is claimed. The initial failure, diagnosis and final checks account for 128 excluded synthetic forwards.

Error detection targets an incorrect two-call mean answer. The combined score averages within-stratum midranks of normalized mean-distribution entropy and TV, $\frac{1}{2} \sum_i |p_i - q_i|$. Its entropy-only reference also costs two calls. For batch allocation, both pay two calls per question, then complete five-call means for their top 80 items: $2(240) + 3(80) = 720$. Gold and future members never enter selection. The extension’s primary stratum is Phi-English; other strata are secondary. Tasks and unequal model sizes preclude causal language/family attribution.

Both packages retain two error targets, eight scores and four policies at five budgets. The 2,000 paired-question percentile bootstraps condition on observed ranks/allocations, with unad-

Checkpoint/task	Errors	AP difference [95% interval]
Q1.5 / old JA	52	-0.024 [-0.092, 0.067]
Q35 / old JA	5	-0.102 [-0.464, 0.124]
Q1.5 / new JA	40	-0.048 [-0.151, 0.039]
Phi3.8 / new JA	37	-0.067 [-0.141, 0.014]
Q1.5 / EN	74	-0.032 [-0.082, 0.022]
Phi3.8 / EN	62	-0.051 [-0.119, 0.013]

Table 5: Entropy-plus-TV minus entropy AP; 240 questions per row. Old JA is the earlier panel. New JA and EN are separately shared across the extension models, not paired translations. Q denotes Qwen; Phi is 3.8B.

justed descriptive intervals and undefined counts retained. Collection pays all five calls; replay costs sum the chosen physical members. Recorded token/latency sums are not online timing or speedups. Reanalysis is not independent inference or human validation.

Aggregation-only control. A [post-hoc reanalysis](#) compares arithmetic means with $g_k \propto \exp(\frac{1}{5} \sum_r \log p_{rk})$ on the same five extension responses. Fixed token–meaning bindings and $T = 1$ remain unchanged; this does not reproduce AnyJev’s position-bound labels or prior correction. All four strata were already inspected; a local plan recorded before this calculation is not preregistration. Table 6 reports all strata. Arithmetic and geometric means share 1,200 physical calls per row; the first-order reference uses 240. No new inference or equal-budget advantage over that reference is claimed. Intervals use 10,000 paired-question bootstrap draws, exploratory and unadjusted; no consistent aggregation winner is established.

Stratum	Correct F/A/G	G–A pp [95% interval]
Q1.5 / JA	194/200/198	-0.83 [-2.50, 0.83]
Q1.5 / EN	166/170/170	0.00 [-1.67, 1.67]
Phi3.8 / JA	204/205/202	-1.25 [-2.92, 0.00]
Phi3.8 / EN	180/174/178	1.67 [-0.42, 4.17]

Table 6: Correct counts among 240 items per row: first-order (F), arithmetic (A), geometric (G). Models share the 480 distinct questions; strata are not pooled.

Human evaluation boundary. No human study was conducted. Synthetic checks and a Claude Code rehearsal are software QA. A prepared importer-versus-complete-JSON pilot has zero participants. Human comparative performance and independent-machine inference remain open.